

Philosophy of Science / Videnskabelig Metode Spring exam 2023

This document provides insight into how exam answers were graded. You can hopefully use it to make sense of the grade you received. Note, it does not provide an exhaustive list, but illustrates typical mistakes as well as what generally characterizes a good answer. First some general comments, and then I will address each question individually.

- Conveying [=give] understanding of the curriculum and using the curriculum to answer the question is the basis for grading the exams.
- Reasons should be provided in each answer. E.g. when assessing question #2 it is not enough to simply write that we need more information on IT expertise and logistics. What information, more specifically, would you have liked to know? Similarly, in question #3 you were asked to explain similarities and differences *in detail*. Stating that post-positivism and constructivism entail different ontological assumptions is not adequate.
- Writing a couple of sentences (in addition to quotes) is usually not enough to constitute a “detailed explanation.”
- Sometimes the question is only partially answered. For example, in question #1 the weaknesses of the studies might have been explained, but how the studies can remedy [=afhjælpe] weaknesses of the others might have been ignored. Similar, in question #3 the similarities of post-positivism and constructivism might have been explained, but not the differences.
- The grade is given based on an overall, holistic assessment.
- On average, question #1 and #3 generated the most points, while question 4 scored the least. The average this year is fairly similar to last year.
- The re-exam is expected to take place in the beginning of August. I will offer a lecture where I, in detail, discuss each question, either in late July or early August. I hope to deliver it onsite, but it might be online.

I might update this document over the weekend to specify some points, in response to questions. [This will be noted here and done in a blue font.](#)

I would like to get permission to use some answers as good, illustrative examples. Flow id's that I would like to use as good examples and upload (parts of) to Brightspace: (to be updated, when grades are given)

Videnskabelig Metode, flow id's: 285, 374, 375, 399, 401 (jeg har brug for jeres tilladelse til at bruge opgaven - hvis I vil give den, må I gerne skrive til mig)

Question 1 (25%)

A passed answer should convey a basic understanding of the weaknesses of the three studies, and explain how a given study's weakness can be remedied by the other studies. More specifically, the following points should be addressed:

- Cross-sectional studies imply the risk of confounding variables, or omitted variable bias (or similar terminology). It is not adequate to write the generic statement that cross-sectional studies have low internal validity. One should specify the given weakness (e.g. that even though control variables are included in study 1 / 2, some

variables might be confounding (or maybe there are variables missing, constituting an omitted variable bias).

- The RCT (study 3) can eliminate the risk of such confounding variables/omitted variable bias (in study 1-2), but implies lower external validity.
- Study 1 + 2 complement each other and imply higher external validity compared to 3.

Good answers note some of the following points:

- That study 1 and 2 are carried out in slightly different economic contexts (high growth, neutral, recession), thus increasing the validity of the claim (both internal and external). They are also in different national/cultural contexts, and still get the same result, thus improving the validity of the claim.
- Relatedly, study 1 wasn't carried out in a recession, and one can argue that the research design doesn't quite match the research question / hypothesis. One could also talk about "inappropriate samples" (as AMJ 2011) does.
- Overall external validity seems strong, given the range of samples, time in history data is collected and study methods used (regular self-perceived survey + RCT)
- While some control variables are collected (in study 1 and 2), others might be important. Good answers specify what variables might be of interest, and why a given variable might both explain the independent and dependent variable (thus being a confounding). For example, life in the cities might be related to a better economic outlook, but also a more stressful work life, thus making it appear that economic outlook is related to job satisfaction, while it in fact is a matter of urban vs. rural life.
- Common methods bias/variance: The first two studies collect data on both the dependent and independent variable from the survey respondents. While it is the perception of the individual that researchers are interested in (rather than the factual economic conditions), this does constitute a potential issue.
- Distinction between omitted and confounding variables.
- This is not an exhaustive list

A few further notes:

- A challenge for some exams was that they might have identified weaknesses of the studies, but didn't explain how the weaknesses of the three studies can be remedied [=afhjulpet] by the other given studies.
- Study 3 should not be considered a quasi-experiment. It compares a good economy with a bad economy. This implies we don't know if the effect is from the bad economy, or the good economy. Yet, this is an external validity problem (as explained in relation to the power pose studies in the lectures). It does not mean the study is not a randomized controlled trial. One is changing from a negative to a positive, or from a positive to a negative perception. If the economy was in fact negative at the time, it would also, in some sense, imply a change if the control group was neutral. This issue also relates to how one in a study of incentives might compare giving 2 DKK to 10 DKK per study; this implies one change, but the control group was not 0 DKK.
- Finally, the first two studies should not be categorized as field experiments, survey experiments or quasi-experiments.

The questions are inspired by this very recent paper, published in the journal "AMJ":
<https://journals.aom.org/doi/10.5465/amj.2019.0405>

Question 2 (15%)

An answer should convey that while there was an allocation to treatment or condition, the process was not based on a randomized process. Yet, might it have been "as good as random"? We therefore need to know more about a) the allocation to treatment, b) if the units in the different conditions really are as similar as possible, and how c) if there really was just one treatment. Recall that a quasi-experiment is a setup that doesn't quite fulfill the criteria for a RCT - and the prior a/b/c criteria are exactly those. Also note that the text in Q2 states that IT expertise and logistics were **partial** reasons for the distribution of units to treatment and control. What other reasons were relevant?

Good answers will for example:

- Explain how the lack of randomization introduces the risk of confounding factors or selection bias.
 - Selection bias: Maybe the availability of IT expertise means that the most IT knowledgeable units got the treatment, which challenges internal validity (selection bias), since more IT knowledgeable units might be better at using the chatbot. Thus, we need to know more about this risk.
 - Confounding variables: One could ask for more control variables, e.g. noting that the study did not control for any employee skills or variance in market conditions across units, or an assessment of the pre-treatment performance of units that got access to a chatbot. Thus, we need to know more about this risk.
- Refer to examples used in class, in particular a study on how test times impact performance of elementary school kids (a quasi-RCT) or maybe a study on mask mandates in Kansas (not a quasi-RCT).
- Explain in detail how confounding variables might threaten the internal validity. Maybe those with IT expertise are located in certain geographic regions, or in areas where sales went up anyways (but due to disconnected reasons) etc.
- Maybe a Hawthorne effect is in play: The treatment group might be happy about being selected, while the control group might be unhappy / jealous about being ignored. This would in particular be a problem if the management team was involved in visiting the units that are in the treatment group.

An answer shouldn't just state that while it wasn't a random process, we just need more information on if IT expertise and logistical challenges constitute "as good as random" factors (too unspecific). Or, that we need more information on the availability of IT expertise; such a question should be linked to the categorization of quasi-experiments.

Question 3 (20%)

In order to pass, an answer needs to a) provide at least two quotes and explain what they mean, b) somehow cover and explain that the positions epistemologically speaking have some similarities, and c) address and explain their differences. Note, it is not adequate to

simply state that they have different perspectives on, for example, objectivity, but one also needs to explain the difference in detail.

Quite a few answers clearly rely on pre-prepared presentation of for example post-positivism. That is of course perfectly ok, but it is important that the answer conveys genuine understanding of the similarities/differences between the two positions. Some answers are fairly long, but the actual comparison of differences/similarities might just be 4-5 lines. Furthermore, a core challenge in many exams relates to a flawed presentation of core concepts. Maybe the overall difference is (correctly) stated, but the detailed explanation of the difference might be wrong.

Relevant points one could make:

Similarities

- This is probably the most relevant quote and will be provided by many: “Even post-positivists have conceded that objectivity is not possible, the results of an inquiry are always shaped by the interaction of inquirer and inquired into” (both PP & C)
- Some will refer to a clip of “Yes Prime Minister” that I showed in class. It illustrates a problem that both PP and C agree we have (neutral observations of the world not possible). They have different solutions to the problem though.
- Complete Truths with capital T is unachievable, due to human limitations, biases. This similarity leads into differences (either modified objectivity, or subjectivity).
- Both agree qualitative methods and multiple perspectives could be useful

Differences

- PP assumes that there are regularities to uncover, and seeks approximate truths to uncover those $\neq$ C
- PP argues we should aim to reduce biases, by relying on triangulation, distanced
- C: “...it makes the findings of an inquiry not a report about what is “out there” but the residue of a process that literally creates them[findings]” $\neq$ PP
- PP: “...although a real world driven by real natural causes exist, it is impossible for humans truly to perceive it with their imperfect sensory and intellectual mechanisms” $\neq$ C
- C argues that realities are multiple $\neq$ PP
- C: “If realities exist only in respondents’ minds, subjective interaction seems to be the only way to access them” $\neq$ PP
- C: “Subjectivity is not only forced on us by the human condition (as the post-positivists might admit) but because it is the only means of unlocking the constructions held by individuals.” (Guba 1990, p.26)

Question 4 (15%)

Please note that the questions asked you to identify which featureS are missing from the given description.

The exam set provides some information that is relevant, such as: interconnectedness, analyzing various elements, deep understanding of dynamics of the system, very difficult

systems etc. This does not fully capture the essence of complexity though. Notice in particular the text says: “A complex system or organisation is characterised by a large number of **interconnected** elements or variables that relate to each other in often **complicated** ways.” Main elements of the following insights (see bullet points) should be part of an answer that can pass. While the insights in the points are connected, a passed answer does not need to mention them all.

- **Interdependence**: Elements/components in a system interact and adapt to each other. This constitutes feedback loops and change in the behavior that can be difficult to predict (-> complexity, see examples with ants, Yellowstone, the three-body problem etc.). The text in question 4 solely talks about elements being interconnected but not how one needs to study the interdependent relations and how they are key to understanding a complex system.
- **The whole is more than the sum of its parts (emergence)**: Even if one knows all the parts in detail, one can't predict behavior, since one needs to know how parts interact/are interdependent on each other's behavior
- **Watts quote**: “It is hard to understand, for example, why even a single organization behaves the way it does without considering the individuals who work in it the other organizations with which it competes, cooperates and compares itself to the institutional and regulatory structure within which it operates the interactions between all these components.” (p. 489). **Focus should be on “interactions between ... components” angle, explaining what this means.**

Could also mention additional points:

- Sargut (from curriculum) defines complexity as: Multiplicity, interdependence, diversity. Notice, just mentioning these terms is not enough - they should be explained. Diversity is clearly not referred to in the text in Q4, neither is interdependence.
- Feldman-Barrett, about mechanistic vs. complexity mindset
- The description in Q4 describes something that is complicated, rather than complex.
- Small changes to initial conditions can have big impacts
- Non-linearity

Nelson or Aschwanden should not be used to make the complexity point (since they don't address feedback loops and interdependence). Furthermore, some argue that the human factor is key, however, as illustrated in class, complexity can be found among ants, animals and all sorts of systems where each individual element is quite simple.

Question 5 (25%)

This is a fairly difficult, open-ended question and an answer could go in many directions. The aim is that they should use curriculum insight to think critically about what they think a meta-analysis is. The questions ask for insight**S** and mentions challenge**S**, so any answer needs to take in multiple perspectives and/or arguments.

Possible angles on why it might be a powerful tool, roughly in order of most relevance:

- Aschwanden: Different researchers have different biases and make different decisions, and if the endpoint is still reasonably similar despite such differences, results are more trustworthy and 'closer to the truth'. Note that some answers merely state that Aschwanden highlights the importance of multiple studies. A better answer explains why Aschwanden's point links to the essence of a meta-analysis (balancing out different biases and subjective choices made by different researchers).
- External validity is likely enhanced, since studies from different sources are most likely included. Similarly, internal validity is likely to be enhanced, since the use of different datasets, and analytical choices might reduce or at least identify the possible impact of variables that are omitted in some studies, thus also helping to identify which variables are "merely" confounding.
- Kahneman: Researchers have biases, and therefore important not just to rely on one or few perspectives. A collective of perspectives (biases) can provide a better picture.
- Nelson: Variables are heterogeneous and influenced by many different variables. This can be considered an advantage, but also potentially a challenge, if variables are too different and potentially incomparable.
- Post-positivism: Similar to Aschwanden. Multiple perspectives, triangulation (of data, researchers, methods etc.).
- AMJ 2011: Need for doing different, multiple studies. The authors of AMJ 2011 don't really have meta-analyses in mind, but the gist of the use of AMJ 2011 in this context is not wrong per se. However, note that the presentation of a meta-analysis refers to a quantitative, statistical analysis. Referring to case studies thus not appropriate.
- Minor point: Risk of fraud. If more people get to the same conclusion, less risk of fraud shaping results.

Possible angles on what challenges the approach might not overcome:

- Aschwanden: Provides an example of US economy/politics, where one can get significant results, but they are in fact meaningless. Lots of significant studies thus don't necessarily imply that the average is meaningful. One could also refer to the replication crisis that she discusses in her text.
- Nelson: One could make the reverse point of the above. If people rely on different definitions and angles on the phenomena, they might be comparing pears and apples.
- Kahneman: The ones doing the meta-analysis might be biased, leading to a certain (confirmation-bias influenced) outcome.
- General research design/biases: A big biased dataset is not better than a small biased dataset. Further, if all studies relied upon are biased (in the same direction), a collective analysis might not remedy such biases.
- Complexity science: This position argues that even very similar RCT settings, might involve slightly different variables and setups, which could lead to interdependence/change in behavior (Feldman-Barret 2021 argument, among others)
- Kuhn: A meta-analysis might just be a reflection of a given paradigm, and thus provide very incremental insights. Thus, a meta-analysis works well for normal science, less so in times where paradigms are being challenged.

Philosophy of Science exam 2022

This document provides insight into how exam answers were graded. You can hopefully use it to make sense of the grade you received. Note, it does not provide an exhaustive list, but illustrates typical mistakes as well as what generally characterizes a good answer. First some general comments, and then I will address each question individually.

- Conveying understanding of the curriculum and using the curriculum to answer the question is the basis for grading the exams.
- Reasons should be provided in each answer. E.g. when assessing question #3 it is not enough to simply write that sentence 5 is not authored by Kahneman. As requested in the question, a detailed explanation should be provided. Similarly, in question #1, it is not enough to simply write: "The external validity seems low because it is a particular lab-setting and they mainly relied on students / people from the Aarhus area".
- Writing a couple of sentences (in addition to quotes) is usually not enough to constitute a "detailed explanation".
- The grade is given based on an overall assessment.
- On average, question 3 and 5 generated the most points, while question 4 scored the least. However, there wasn't a big difference between the questions. The average this year is fairly similar to last year, although fewer students failed.
- The re-exam is expected to take place in the beginning of August. I will offer a lecture where I, in detail, discuss each question, probably in late July. I hope to deliver it onsite, but it might be online.

I might update this document over the weekend to specify some points, in response to questions. [This will be noted here and done in a blue font.](#)

I would like to get permission to use some answers as good, illustrative examples. Flow id's that I would like to use as good examples and upload (parts of) to Brightspace:

Philosophy of Science: 131, 94

Question 1 (20%)

A) Please explain in detail what the internal and external validity of the presented research design is.

Internal validity

- A passed answer should be able to identify that (and why) this is a randomized controlled trial, thus the internal validity (IV) can be argued to be high.
- Note that even though people have self-selected to participate (in a certain time-slot) the process is still randomized. With good reasons it is fair to be somewhat critical of the randomization process (maybe friends showed up in groups), but the exam should also clearly acknowledge the randomization process. Focusing too much on the different time slots usually wasn't helpful. Note the paper is published as a RCT.

External validity

A passed answer should rely on multiple of the following criteria, in some detail.

- Selection and treatment: Only certain kinds of individuals participate and this limits EV, that is the ability to generalize beyond the study. They can be characterized as mainly students, as people from the Aarhus area, and/or as people who have self-selected into participating. Note, that this self-selection angle should not be related to internal validity, since it is an external validity issue.
- Setting and treatment: The tasks might be difficult and reflect relevant skills, but don't represent typical organizational tasks. The fact that there are two tasks (not just one) is positive, good answers will mention this. The payment structure is meaningful in the sense that organizations can adjust pay rates, so this is a relevant organizational lever.
- History and treatment: Less relevant in question a). Not an historical change between settings.
- Hawthorne effect: People are monitored, which might influence behavior in the lab.

This leads to external validity being limited. However, not as low as in Ariely et al. (2008) e.g. (the Bionicle study).

B) Let us hypothetically assume that the above outlined study was also carried out in a poor, rural area in a developing country. The same monetary stakes are used, converted to the local currency. Again no significant effect of the size of the reward on performance is found. How would these findings impact the internal and / or the external validity of the above outlined Holst-Hansen & Bergenholtz (2020) study? Please be specific in your answer.

Multiple of the following points should be made.

Internal validity

- IV strong(er). Bigger sample and repeating the study strengthens belief that there is no relation.

External validity

- Selection: Now broader selection of people so less threat of an interaction between selection and treatment
- Setting: Different (in terms of value of money) setting, so somewhat different interaction of setting and treatment. Strengthens external validity that we get the same result with different stakes.
- History: Not different history as such, but different society, so fair to invoke this argument.
- Hawthorne: Not much information provided so probably the same as in the original study.

Overall, one should argue that the external validity is now stronger.

These points do not constitute an exhaustive list of good / bad elements in exam answers.

Btw., quite a few have assumed and argued that due to illiteracy, the participants in the hypothetical study would not be able to solve the tasks. Yet, the majority of people in

developing countries, even in rural areas, are not illiterate. Furthermore, the lack of a result (statistical insignificance) does not (negatively) impact internal validity.

Question 2 (20%)

i) Research design

Three research designs could be identified.

- Longitudinal since data from several years were collected, and e.g. used in the assessment of the role of the salary increase. This was not a quasi-experiment.
- Cross-sectional, since (also) relying on non-longitudinal data and focus on comparing (variation in) performance of night vs. day-shift.
- Case study, since focus on an in-depth investigation of group dynamics.

A passed answer need not have identified all three, but at least two of the elements, and be able to describe the nature of them.

Possible weaknesses of the research designs

- Possible omitted variable bias (in particular in relation to the difference between night and day environments)
- Possible poor measurement since different managers rated 'own' employees, which could lead to measurement / construct bias
- Manager both assess employee performance as well as shift performance, related to common methods bias
- Qualitative study weakness: Possible selection bias, subjectivity / confirmation bias

Answers should not be generic (about research designs in general) or be written without any terminology/insights from the curriculum and course material. AMJ (2011) is a core reference to rely on, although other sources are useful as well (BRM e.g.).

Core challenge for many exams here is not clearly relying on the curriculum / course material to answer the question.

ii) Certainty

The manager should not be certain that day-shift employees would perform better. The main points to rely on were addresses above, and should constitute the starting point of the answer here.

Question 3 (16%)

Which of the following sentences is Kahneman unlikely to have authored? Please be specific in your explanation of why some of the sentences do not match Kahneman's presentation of system 1 vs. system 2.

Bemærk den danske udgave af spørgsmål kunne være tvetydig: svar på baseret på "hvilke" og "hvilken" blev begge godkendt. Idet kun en sætning var entydigt forkert, fungerede denne tilgang.

Of the sentences, #5 is clearly wrong, while sentence #2 and part of sentence #3 are at least questionable.

- 1) *System 1 is effective but also implies a range of cognitive systematic biases.*
 - Kahneman would agree, this is a core feature of his explanation of the two systems.
- 2) *My message is that it is possible to greatly reduce the number of these systematic biases.*
 - Kahneman would probably object to the word 'greatly'. Biases can be reduced by being aware of the systems and one's own particular biases, but only to a limited degree.
- 3) *System 2 is a mental process that requires effort, but an individual can ensure to activate system 2 when necessary, thus reducing the number of systematically biased errors system 1 might make.*
 - System 2 is in fact a mental process that requires effort but one can not 'ensure' to activate system 2 when necessary. See quotes below.
- 4) *Thus, the more one is aware of the systematic mistakes one tends to make, the better chance one will have to limit the number of biases to which one falls prey.*
 - Kahneman would probably agree with this sentence. If one is aware of biases (e.g. confirmation or anchoring bias) one has an opportunity to act against them.
- 5) *Interestingly, since one knows oneself best, it is easier to identify the mistakes oneself makes, rather than the biases to which one's friends and colleagues fall prey.*
 - Since one is relying on a biased system, one is less likely to identify one's own biases. Actually, Kahneman makes the exact opposite statement (p. 28), cf. the quote below.

Again, answers should explain why the claimed quote can't be attributed to Kahneman - remember, the exam questions asked for a detailed explanation. The key focus when assessing the answer is if the answer demonstrates understanding of Kahneman's work. A passed answer should be able to identify issues with at least two of the sentences.

Some relevant quotes one could rely on are:

- *"Constantly questioning our own thinking would be impossibly tedious, and System 2 is much too slow and inefficient to serve as a substitute for System 1 in making routine decisions." (p. 28)*
- *"Because System 1 operates automatically and cannot be turned off at will, errors of intuitive thought are often difficult to prevent. Biases cannot always be avoided, because System 2 may have no clue to the error. Even when cues to likely errors are available, errors can be prevented only by the enhanced monitoring and effortful activity of System 2. As a way to live your life, however, continuous vigilance is not necessarily good, and it is certainly impractical." (p. 28)*
- *"The best we can do is a compromise: learn to recognize situations in which mistakes are likely and try harder to avoid significant mistakes when the stakes are high. The premise of this book is that it is easier to recognize other people's mistakes than our own." (p. 28)*

Question 4 (16%)

A fairly open-ended question where several approaches can be used.

Nelson 2016

For one, Nelson argues that it is difficult to measure relevant constructs and qualitative proxies have to be made. Boundaries of constructs are fuzzy and many variables influence the outcomes we are interested in. Thus, results are somewhat uncertain and depend on choices made.

Second, the subject matter might change, as shown in this quote:

- *“the social sciences also have to face the problem that the subject matter they study often changes over time, and hence both the entities they study and the basic causal relationships affecting these variables may be different today than they were last year, or a decade ago”* (Nelson 2016, p. 1698)

Thus, different results could be due to this change in behavior over time, in principle.

Bergenholtz & Busch 2016

Making theories public might change behavior due to the awareness of the theoretical mechanism. Thus, social science theories can in principle be self-fulfilling or self-falsifying (as illustrated by the basketball-penalty examples in the text and discussed in class. An answer could thus rely on this text and perspective, and should explain it in further detail.

Aschwanden 2015

Aschwanden's text shows how different groups of researchers might reach different conclusions, even if they are competent and non-biased. Science is complicated and it is difficult to decide which variables are relevant. Thus, the original results might simply reflect the variance of results we can expect when analyzing datasets - especially different ones. See e.g. a quote like this: "Even the most skilled researchers must make subjective choices that have a huge impact on the result they find." To show that the original study might not be false, one could argue that the later study might be more biased or the fact that even more studies are needed to get the full picture (cf. the football-red card example in Aschwanden's text).

Kuhn

In principle, paradigmatic changes might imply that making comparative assessments of the truth of theories across decades is too difficult. Applying Kuhn then requires explaining Kuhn's concept of paradigm changes.

Constructivism (Guba)

Multiple realities arguments.

Other approaches are probably also possible (e.g. a version of interaction of treatment and history, BRM), I've only mentioned the most obvious ones here.

Question 5 (16%)

Guba (1990) proposes categories to capture different ontological, epistemological and methodological assumptions. In our class and curriculum we dealt with three categories; positivism, post-positivism and constructivism. Please explain in detail which of these three categorisations Kuhn's position is the most similar to (if any). Provide quotes that support your arguments, from both Guba and Kuhn (at least one relating to each).

This is a relatively difficult question, but I have, briefly, provided a crude outline of an answer in class, where I posed this question.

Positivism: Kuhn's position is obviously not at all similar to positivism, due to ontological and epistemological reasons. This quote makes this point clearly: "Kuhn p. 206: There is, I think, no theory-independent way to reconstruct phrases like 'really there'; the notion of a match between the ontology of a theory and its "real" counterpart in nature now seems to me illusive in principle".

Post-positivism: Since this position also implies some ontological notion of the theoretical mechanism / construct being "really there" (although epistemologically not fully accessible), it constitutes a mis-match to Kuhn's arguments. However, the position also implies that science will improve, albeit never reaching any endpoint - I will return to this below. Some students have used the "electron" quote (p. 196 in Kuhn) to argue that Kuhn thinks electrons exist out there, in some independent fashion, and we are trying to measure this identifiable external world. This is not a correct reading of Kuhn.

Constructivism: This is the position that Kuhn should primarily be associated with. There is no ontologically independent 'real world' out there that we are trying to uncover, and epistemologically speaking there is no theory-independent way to access the world. Theories are constructed in some sense.

However, Kuhn did try to argue against the relativism that constructivism implies, by arguing that science does improve over time: Not by resembling or getting closer to the truth, but via an evolutionary perspective: later paradigms tend to be better at solving scientific puzzles. In this sense one could argue that there are elements of post-positivism in Kuhn (or that it is difficult to exactly pinpoint Kuhn's position) - however, it is important to distinguish this from post-positivism's perspective on truth.

Overall, merely identifying constructivism is not adequate, a detailed explanation has to be provided as well as relevant quotes. Furthermore, if post-positivism is invoked, the answer needs to make clear that Kuhn is not agreeing on getting closer to the Truth.

Question 6 (12%)

Is the overall message provided in Feldman-Barrett (2021) the most similar to elements in Nelson (2016), Aschwanden (2015) or Kuhn (1962)? Please provide a detailed explanation for your choice.

It is the easiest to relate FB 2021 to **Nelson (2016)**. The core message of FB 2021 is that researchers often underestimate how many weak, interacting factors might influence the dependent variable (complexity vs. mechanistic mindset). This should be explained in some (further) detail.

A relevant quote from Nelson also shown in class:

- *“Second, the forces and conditions that influence the subjects of study in ways that the science seeks to understand are numerous, highly variable, and often cannot be separated sharply one from another. Indeed the circumstances associated with any particular set of observations may need to be understood as, in some sense, unique.”*
(p. 1697)

Nelson doesn't use the (technical) concept of complexity as such, but makes arguments in a similar spirit. See also this quote by Nelson: "I propose that many of the most pressing problems that society today wants science to illuminate involve complex phenomena that vary significantly from instance to instance and that are influenced by many factors that also often vary greatly." An answer should outline Nelson's position in detail, and relate it to FB 2021.

One could relate FB 2021 to **Aschwanden**. However, Aschwanden doesn't talk about complexity, but about how many different factors can impact the dependent variable in unknown and difficult to identify ways. So, doesn't really talk about how they "interact" (cf. complexity). One could also link FB 2021 to Aschwanden via their focus on the replication crisis, yet this is not the primary message of FB 2021 though.

It is not straightforward to relate FB 2021 to **Kuhn**. However, one could make a meta-argument, stating that Kuhn would say that mechanistic vs. complexity mindset (presented in FB 2021) constitute two different paradigms.